MIL-H-9697(USAF)
6 August 1958

MILITARY SPECIFICATION

HEADSET-MICROPHONE H-163( )/U

1. SCOPE

1.1 This specification covers one type of headset-microphone and cord, designated Headset-Microphone H-163( )/U, hereinafter referred to as "the headset-microphone".

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitations for bids, form a part of this specification:

SPECIFICATIONS

Federal

QQ-A-318	Aluminum Alloy 52S; Plate and Sheet

Military

MIL-P-642	Plugs, Telephone
MIL-C-3849	Cord, Electrical (Tinsel)
MIL-E-4158	Electronic Equipment, Ground; General Requirements For
MIL-T-4807	Tests; Vibration and Shock Ground Electronic Equipment (Requirements For)
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification For
MIL-T-9107	Test Reports; Preparation Of
MIL-P-17555	Preparation For Delivery of Electronic Equipment; Miscellaneous Electrical Equipment (Except Rotating Electrical Equipment) and Associated Repair Parts

DRAWINGS

Air Force

56K12858	Cushion, Airfoam MX-2088/U Assembly Of

(Copies of applicable documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC-5965

THE REPUBLICAN PRESS, Hamilton, Ohio
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2.2 OTHER PUBLICATIONS.- The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Consolidated Classification Committee
Consolidated Freight Classification Rules

(Applications for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Illinois).

American Standards Association
Z24.4/164 Pressure Calibration of Laboratory Standard Pressure Microphones, Method for the

(Copies of ASA Standards may be obtained from the American Standards Association, Incorporated, 70 East 45th Street, New York 17, New York).

3. REQUIREMENTS

3.1 PREPRODUCTION.- This specification makes provisions for preproduction testing.

3.2 COMPONENTS.- The headset-microphone shall consist of:

Item No.	Quantity	Description	Requirement
1	2 each	Receiver Housing	3.6.1
2	1 each	Microphone Housing	3.6.2
3	1 each	Headband Assembly	3.6.3
4	1 each	Cord	3.6.4

3.3 GENERAL SPECIFICATION.- The requirements of MIL-E-4158 apply as requirements of this specification. Where the requirements of the general specification and this specification conflict, the requirements of this specification shall govern. Exceptions and additions to the general specification shall be as follows.

3.3.1 AMBIENT TEMPERATURE.- The headset-microphone shall be designed to operate in a "Cold Weather Area" and in "Desert and Tropical Areas".

3.3.2 SERVICE CONDITION ([redacted]).- The headset-microphone shall not suffer damage nor fail to give the required performance when subjected to conditions encountered during operation, storage and transit.

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3.3.3 SERVICE CONDITION (ELECTRICAL).- The headset-microphone shall be designed to operate from a 24 plus or minus 4 volts, direct-current (d-c) power source.

3.3.4 SERVICE LIFE.- The headset-microphone shall be capable of operating 24 hours per day for a period of 2 years with only normal maintenance and without major overhaul. The operational period may start after an initial storage period not exceeding 2 years.

3.3.5 CORD ELECTRICAL.- The three-conductor tinsel cord shall be in accordance with MIL-C-3849.

3.3.6 INSULATION MATERIAL.- Insulation material shall not support rapid combustion. The material used shall be sufficiently rigid and free from cold flow to maintain its function.

3.3.7 ALUMINUM.- Aluminum shall be in accordance with QQ-A-318.

3.3.8 QUALIFICATION OF WELDERS.- Qualification of welders shall not apply.

3.4 DESIGN AND CONSTRUCTION.- The headset-microphone shall consist of a headband assembly and a microphone. The headband assembly shall have two receiver housings. These housings shall be attached to the headband by means of a movable metal band with an adjustment block and a metal fork. The microphone, an assembly of the receivers housing, shall consist of a hollow square boom anchored inside the receivers housing on one end, and a transmitter housing anchored on the other end. Both receivers and the transmitter housing shall each have a screw cap for holding the elements in place. The square boom shall contain the wiring between the transmitter and receivers elements with the wiring connected to terminals in the transmitter and receiver units.

3.4.1 RECEIVERS ELEMENT.- The headset-microphone receivers element shall be Western Electric Co. Type HC-4, or equal.

3.4.2 TRANSMITTER ELEMENT.- The headset-microphone transmitter element shall be a carbon type, Western Electric Co. Type N-1 microphone, or equal.

3.5 PERFORMANCE

3.5.1 RECEIVERS ELEMENT

3.5.1.1 IMPEDANCE.- The impedance of the receivers element at 1000 cycles per second (cps) shall be between 250 and 300 ohms.

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3.5.1.2 LEAKAGE RESISTANCE.- Insulation between the case and terminals shall be provided. The insulation shall be capable of withstanding test requirements specified in 4.4.5b without breakdown or decrease of the insulation resistance below 1 megohm between these points.

3.5.1.3 DISTORTION.- The acoustic output of the receivers element shall be free from serious objectionable distortion over the audio frequency range of 200 to 500 cps, as judged by ear.

3.5.1.4 OVER-VOLTAGE.- The response of the receivers element shall show not more than 3 db change from the original response value.

3.5.1.5 FREQUENCY RESPONSE.- The output shall be not lower than the minimum limits shown on Figure 1. In addition, the response shall fall within the envelope as shown on Figure 2.

3.5.2 MICROPHONE ELEMENT

3.5.2.1 IMPEDANCE.- The microphone element shall have an impedance between 15 and 30 ohms at 1000 cps.

3.5.2.2 UNAGITATED CURRENT.- The unagitated current shall not exceed 130 milliamperes.

3.5.2.3 FREQUENCY RESPONSE.- The frequency response of the microphone element shall be within the limits of Figure 3 when measured in the vertical position. It shall be essentially smooth and devoid of any pronounced dips or peaks. For any bandwidth for the various frequency ranges indicated below, the change in response from the lowest to the highest frequency in the band shall not exceed the following limits:

Frequency Range (cps)	Bandwidth (cps)	Change in response (decibels (db))
300 to 1000	100	plus 3.5 to minus 1.0
1000 to 2000	300	plus 5.0 to minus 2.0
2000 to 3000	300	plus 5.0 to minus 4.0
3000 to 4000	500	plus 5.0 to minus 5.0

3.5.2.3.1 The response of the microphone element, at each test frequency in each of the other three test positions, shall not vary more than 5 db from response in the vertical position.

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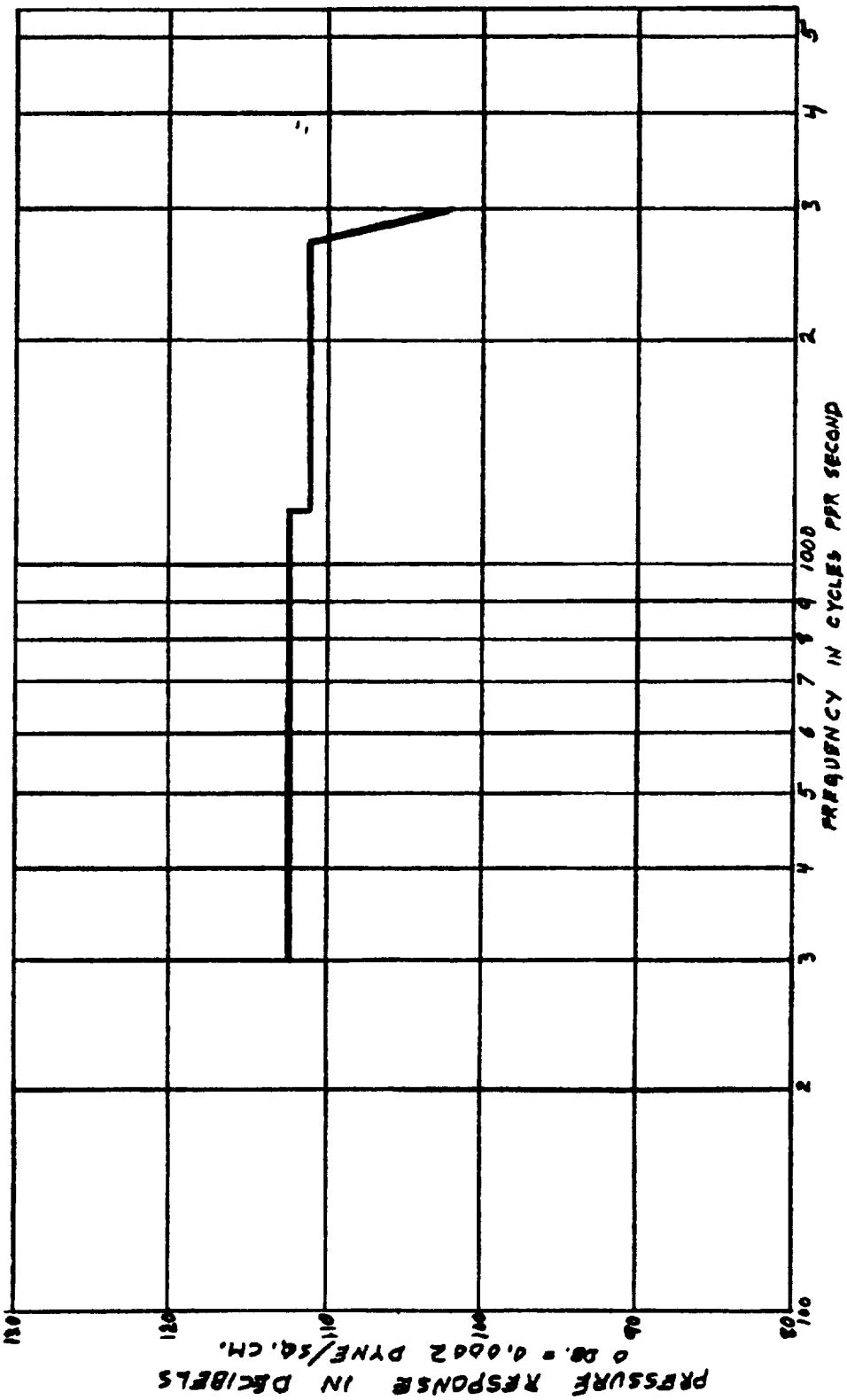


FIGURE 1. MINIMUM RESPONSE FOR RECEIVER ELEMENT

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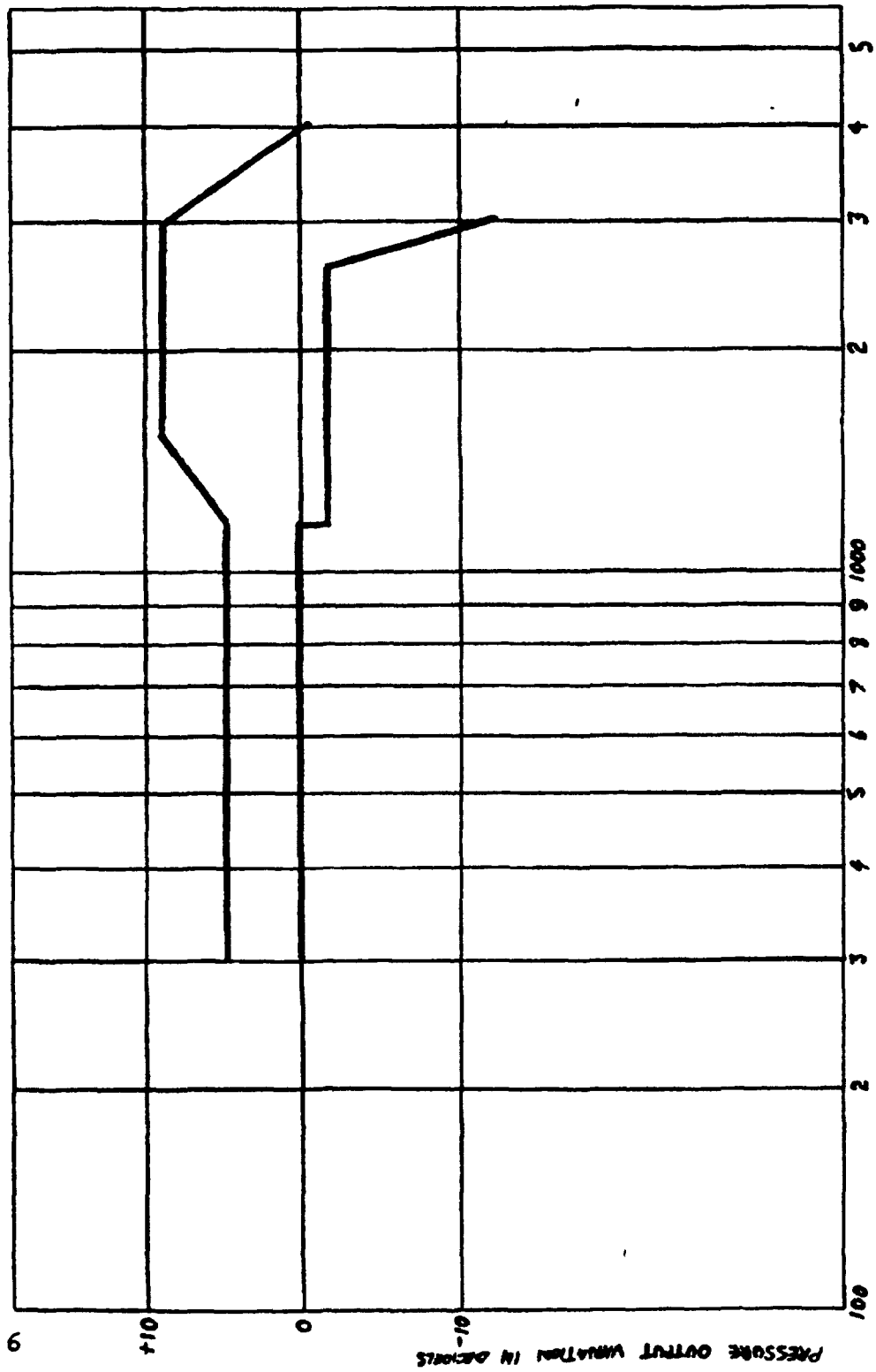


FIGURE 2. RATE OF CHANGE ENVELOPE FOR RECEIVER ELEMENT

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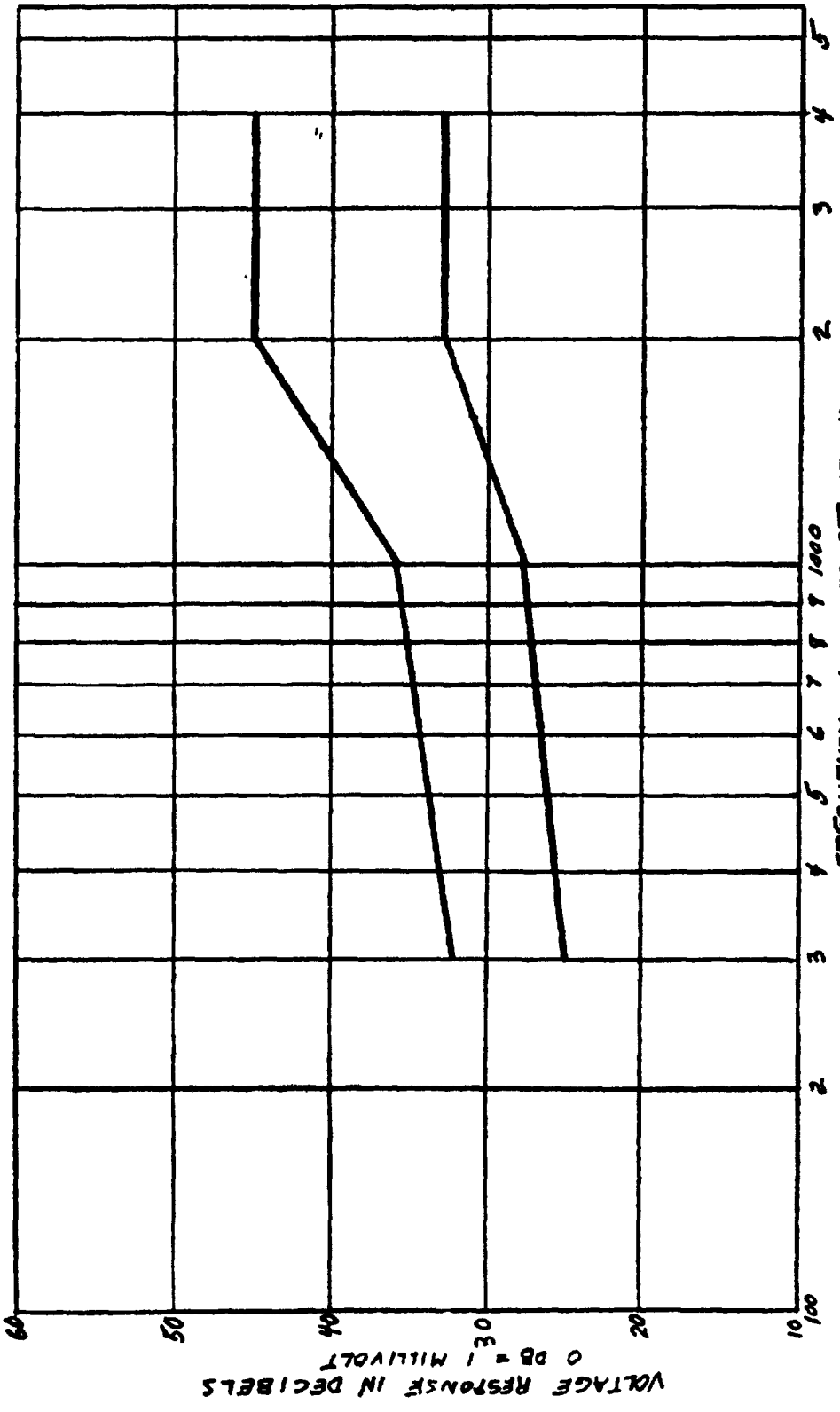


FIGURE 3. RESPONSE LIMITS FOR MICROPHONE ELEMENTS

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3.5.2.4 LOW VOLTAGE.- The response of the microphone element shall be not less than the limits shown below.

Frequency (cps)	Response (decibels referred to 1 Milliwatt (dbm))
300	27
500	28
1000	29
2000	37
3000	40
4000	40

3.6 DETAILS OF COMPONENTS

3.6.1 RECEIVER HOUSING.- The circular slot which accepts the rotating headband assembly shall include a series of detents to permit the microphone to remain in a desired fixed station.

3.6.1.1 MAGNETIC CIRCUIT.- Magnetization of the permanent magnet shall be accomplished by saturation following which the magnet shall be stabilized by reducing the magnetization to that required by the particular design.

3.6.2 MICROPHONE HOUSING.- The microphone housing shall be constructed of a rigid thermoplastic material. It shall consist of a screw type cap for securing the microphone unit and shall be capable of being anchored to the square boom.

3.6.2.1 SQUARE BOOM.- The square boom shall be fabricated as one continuous piece without intermediate connections such as welds, screw connections, et cetera.

3.6.3 HEADBAND ASSEMBLY.- The headband shall be made of spring band covered with a material which shall give a maximum degree of traction. The headband shall accommodate two ear cushions similar to Cushion, Earphone MX-2088/U, Hyperlon type in accordance with drawing 56R-12858.

3.6.3.1 TENSION.- There shall be sufficient tension in the headband assembly to hold the headset-microphone in a fixed position during prolonged operation. The tension shall not create pressure points which would produce discomfort to the operator. The tension shall be capable of accommodately heads ranging from 5.52 to 6.45 inches in breadth.

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3.6.4 CORD.- A three conductor tinsel cord shall be utilized in accordance with MIL-C-3849. One end shall be connected to the headset-microphone and the other end shall terminate in a Plug PJ-068 in accordance with MIL-P-642 as follows:

a. Tip (black) of Plug PJ-068 to one side of headset receiver.

b. Ring (red) of Plug PJ-068 to one side of microphone.

c. Sleeve (white) of Plug PJ-068 to other side of headset and microphone receiver in series with that connected to the tip. This conductor shall be common to one side of the microphone and one side of the receiver.

d. The length of three conductor cord shall be five feet.

4. QUALITY ASSURENCE PROVISIONS

4.1 CLASSIFICATION OF TESTS.- The inspection and testing of the headset-microphone shall be classified as follows:

a. Acceptance tests.....See 4.2
b. Preproduction testing.....See 4.5

4.2 ACCEPTANCE TESTS.- Acceptance tests shall consist of:

a. Individual tests
b. Sampling plans and tests

4.2.1 INDIVIDUAL TEST.- Each headset-microphone shall be subjected to the following test as described under 4.4 "TEST METHODS" of this specification:

a. Examination of Product (See 4.4.1)
b. Electrical Tests (See 4.4.2)

In addition each headset-microphone shall be subjected to any other tests specified in 4.4 which the procuring activity considers necessary to determine compliance with the requirements of this specification.

4.2.2 SAMPLING PLAN AND TESTS

4.2.2.1 LOT.- A lot shall consist of 50 headset-microphone which have passed the individual tests specified in 4.2.1.

4.2.2.2 SAMPLING PLAN.- One headset-microphone shall be selected at random from each lot and subjected to individual test and the following tests:

a. Response test (See 4.4.5e and 4.4.6.1)
b. Impedance test (See 4.4.5a and 4.4.6.2)

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4.2.2.2.1 REJECTION AND RETEST.- When one or more headset-microphone from a lot fail the specified tests, this is cause for rejection of the lot. Before resubmitting the lot, the contractor shall fully explain to the inspector both the corrections made and the cause of previous rejections.

4.2.3 DEFECTS IN ITEMS ALREADY ACCEPTED.- The investigations of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.3 TEST CONDITIONS.- Unless otherwise specified, all tests required by this specification shall be made at prevailing ambient temperature and humidity conditions.

4.4 TEST METHODS

4.4.1 MECHANICAL INSPECTION TESTS.- The headset-microphone shall be given a thorough mechanical and visual inspection and test to determine that the quality of material and workmanship is in compliance with the requirements of this specification. Particular attention shall be given to the following:

a. Completeness
b. Identification markings
c. The fit of components in their respective positions
d. Finishes
e. Workmanship
f. Solder joints
g. Wire connectors
h. Check of insulation
i. Check fit of threaded parts

4.4.2 ELECTRICAL TESTS.- The headset-microphone shall be given thorough electrical tests considered necessary to determine that all circuits are inherently sound so that over-all performance of the headset-microphone in compliance with the requirements of this specification shall be obtained. The electrical tests shall include but not necessarily be restricted to the following.

4.4.2.1 CONTINUITY.- Each electrical component and each cord and cable shall be given a continuity test if it is wired and connected correctly and that good electrical contact is obtained.

4.4.2.2 OPERATING VOLTAGES.- The operating voltage at all important points, shall be checked for conformance to those shown on the circuit label and schematic drawing. This shall be done with all controls set for normal operation of the equipment.

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4.4.3 ENVIRONMENTAL TESTS.- The headset-microphone shall be subjected to the following environmental conditions in accordance with MIL-E-5272 with exceptions as specified:

a. High temperature test (nonoperating) - procedure I: Maintain the equipment at a temperature of plus 71°C for 4 hours, reduce the temperature to plus 52°C for 4 hours, then operate the equipment and compare results.

b. Low temperature test (nonoperating) - procedure II.

c. Humidity test - procedure II: Duration of test shall be 50 hours.

d. Altitude test:

(1) Operating - procedure I: Duration of test shall be 4 hours.

(2) Nonoperating - repeat procedure I at an absolute pressure of 5.54 inches of mercury (pressure at approximately 40,000 feet above sea level) for 4 hours, but do not operate the equipment. Visually examine the equipment in accordance with the specification.

e. Fungus resistance test - procedure I.

f. Salt spray test - procedure I: Duration of test shall be 100 hours.

g. Sand and dust test - procedure I.

4.4.4 MECHANICAL TESTS

4.4.4.1 VIBRATION TEST.- The headset-microphone shall be subjected to a vibration test in accordance with MIL-T-4807, method 1B.

4.4.4.2 DROP TEST.- The equipment shall be dropped six times at random from a height of 6 feet. Impact at the bottom shall be made on a 3/16-inch horizontal sheet of masonite, or equal, (Brinell hardness No. 11 to No. 12) backed by concrete. The equipment shall transmit and receive intelligibly after the test. There shall be no mechanical failure, such as cracking or breaking of parts. Minor clipping of plastic parts shall not constitute a failure. The equipment shall not suffer degradation in its performance of more than 3 db for the microphone element and 2 db for the receiver element.

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4.4.5 RECEIVER ELEMENT TEST.- Unless otherwise specified, the receiver element shall be in its housing and subjected to the following tests:

a. Impedance - With the receiver element in its housing and located on the 6 coupler calibration (cc) coupler, the impedance shall be measured with 1 volt at 1000 cps applied to the receiver element terminals. This shall be accomplished by either measuring the voltage across and the current through the receiver element with an impedance bridge. The results shall be as specified 3.5.1.1.

b. Leakage resistance - A 500-volt rms, 60-cycle, a-c signal shall be applied between the case and the terminals for a period of 5 seconds. Results shall be as specified in 3.5.1.2.

c. Distortion - A voltage of 2.5 volts in the frequency range of 200 to 500 cps shall be applied across the terminals of the receiver element in series with a 275-ohm resistor. It shall then meet the requirements specified in 3.5.1.3.

d. Over-voltage - Ten volts at 400 cps shall be applied directly across the receiver element terminals for 10 hours. It shall then meet the requirements specified in 3.5.1.4.

e. Frequency response - The response of the receiver element shall be measured in the test circuit shownfr in Figure 4. A Western Electric Company 640 AA Standard pressure microphone, or equal, calibrated according to American Standards Association Specification Z24.4/164 shall be used to measure response of the receiver element. A good seal shall be obtained between the receiver housing and the 6 cc test coupler. Frequency response of the receiver element shall be determined from 300 to 4000 cps at frequencies which indicate the over-all response characteristics and the output level of the receiver element. The "constant available power" input to the receiver element shall be 1 milliwatt. Therefore, the audio oscillator shall be set to give a reading of .1005 volts at each test frequency as measured on the vacuum tube voltmeter (VTVM) No. 1. The response of the receiver element shall be measured on VTVM No. 2 and shall be as specified in 3.5.1.5.

4.4.6 MICROPHONE ELEMENT.- Unless otherwise specified, the microphone element shall be in its housing and subjected to the following test.

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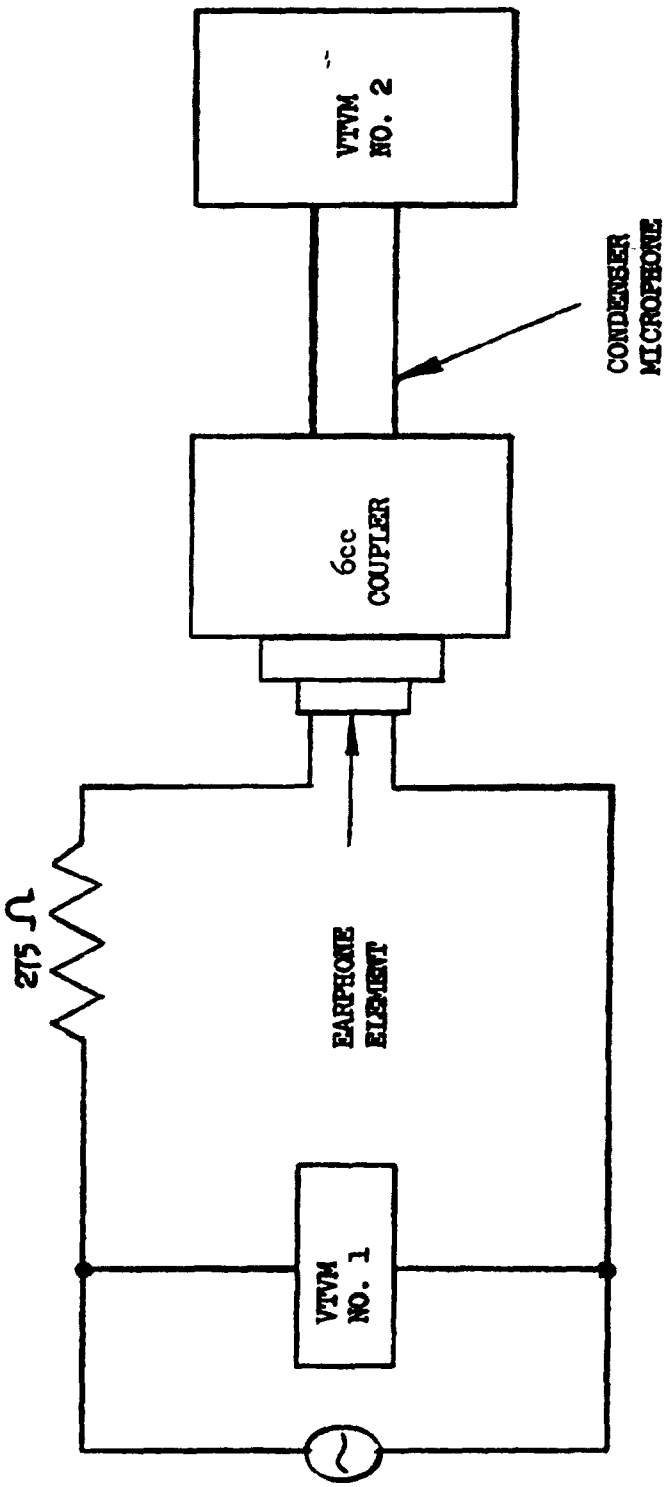


FIGURE 4. TEST CIRCUIT FOR RECEIVER ELEMENT FREQUENCY RESPONSE.

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4.4.6.1 RESPONSE TEST.- Frequency response measurements shall be made every 100 cycles in the range from 300 to 1000 cps and at every 250 cycles in the range from 1000 to 4000 cps. Results shall be as specified in 3.5.2.3.

a. Microphone test circuits - The frequency response of the microphone element shall be measured across a 20-ohm non-inductive load resistance which shall be in series with the microphone element and a 100-microfarad electrolytic condenser. The condenser shall be shunted by a series circuit consisting of a 24-volt d-c source, 250-ohm variable resistance, and a milliammeter. With the microphone element short-circuited, adjust the variable resistance to obtain a d-c current of 85 milliamperes.

b. Voltmeter - The voltage across the load resistance shall be measured with an electronic a-c voltmeter having an input impedance of not less than 0.5 megohm. A Ballantine Model 300, VTVM, or equal, having a flat frequency response from at least 100 to 10,000 cps, shall be used. It shall be capable of measuring voltage from 0.001 to 10 volts rms or more.

c. Sound source - A Western Electric Company No. 555W Artificial Mouth, or equal, shall be used.

d. Audio oscillator - A General Radio type 913B Beat Frequency Oscillator, or equal, having a wave form distortion of less than 2 percent, shall be used.

e. Test apparatus - The test apparatus shall provide mountings for holding and rotating the sound source and microphone element in the following positions:

(1) The microphone element, assembled in its housing, shall be mounted coaxially with the sound source with a 1/4-inch air space between the extreme end of the sound source coupler and the cap of the microphone element housing.

(2) The test apparatus shall permit rotating the sound source and microphone element assembly through 180 degrees in a vertical plane perpendicular to the microphone element diaphram.

f. Sound pressure calibration - A Western Electric 640AA Condenser Microphone, or equal, with its protective grid, shall be used for setting up a sound pressure of .28 dynes per square centimeter. The microphone shall be calibrated by the reciprocity method in accordance with "Calibration of Standard Pressure

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Microphones", American Standard Association Standard Z24.4/164. The plane of the protective grid of the microphone shall be axially located 1/4 inch from the sound source coupler. The electrical input to the sound source shall be adjusted to obtain the specified sound pressure in terms of dynes per square centimeter at each test frequency. The electrical output of the condenser microphone shall be measured either with a volume level indicator or an a-c electronic voltmeter.

g. Conditioning the microphone element prior to test - After calibrating the sound source, replace the condenser microphone with the test microphone element assembled in a suitable housing and with its grid located as specified in 4.4.6.1e. It shall shake the microphone element vigorously in all directions before mounting it in the fixture. With the microphone diaphram in the vertical plane, rotate the microphone element about its axis through an approximately 270-degree arc and back to its original position two times at a regular uniform rate of approximately one complete cycle (270 degrees and back) in 2 seconds. Then rotate the test apparatus to the position in which the microphone element shall be tested. With the current adjusted as specified in 4.4.6.1a and flowing through the microphone element, wait approximately 3 seconds for current stabilization. Then apply an agitation sweep signal at an average sound pressure of 70 dynes per square centimeter and varying uniformly from 1000 to 3000 cps and back to 1000 cps every 2 seconds. At the conclusion of the third sweep, the audio oscillator shall be shut off in such a manner as not to destroy the spurious conditioning.

h. Response measurements - After approximately 3 seconds delay turn on the test tone and, using a sound pressure of 28 dynes per square centimeter, take the response curve starting at 300 cps. Results shall be as specified in 3.5.2.3.

i. Positional characteristics - The frequency response characteristics of the microphone element shall be measured for four microphone positions; with diaphram vertical, with diaphram at 45 degrees (face up), with diaphram face up, and diaphram face down. The microphone element shall be conditioned as specified in 4.4.6.1g prior to each test and then shall meet the requirements specified in 3.5.2.3 and 3.5.2.3.1.

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4.4.6.2 IMPEDANCE TEST.- During the response test specified in 4.4.6.1h, the d-c voltage and d-c current of the microphone element shall be obtained simultaneously at 1000 cps when the microphone element is located in the vertical position. The impedance of the microphone element shall be as specified in 3.5.2.1 and shall be determined by the following formula:

$$Z = V/I$$

Z = Impedance of microphone element
V = d-c voltage across microphone element (at 1000 cps)
I = d-c current through microphone element (at 1000 cps)

4.4.6.3 UNAGITATED CURRENT TEST.- This test shall be performed in a place which shall be relatively free from noise and vibration. The microphone element, in its housing, shall be connected into a test circuit consisting of a 4-1/2-volt d-c source (of negligible impedance) in series with a suitable millimeter and a 20-ohm resistor. The source and the meter shall be shunted by a 100-microfarad electrolyte capacitor. The microphone element shall be rotated as specified in 4.4.6.1g. The circuit shall then be closed and 10 readings of current, at intervals of approximately 1 minute, shall be taken. The highest reading shall be considered as the maximum unagitated current and shall be as specified in 3.5.2.2.

4.4.6.4 LOW VOLTAGE TEST.- The microphone element, assembled in its housing, shall be mounted as specified in 4.4.6.1e (vertical position) and connected as shown in figure 5. Having previously calibrated the sound source for 28 dynes per square centimeter as specified in 4.4.6.1f at 300, 500, 1000, 2000, 3000 and 4000 cps, condition the microphone element as specified in 4.4.6.1g. Turn on the test tone and, using a sound pressure of 28 dynes per square centimeter, take response readings at each of the above frequencies with circuit switch closed. This test shall be completed in 1 minute or less. The response of the microphone element shall meet the requirements specified in 3.5.2.4.

4.4.6.5 THERMAL NOISE AND RESISTANCE.- The test circuit shall be the same as that used for the response test specified in 4.4.6.1 with the current adjusted as specified in 4.4.6.1a. The a-c meter, used to measure the voltage drop in the load resistance caused by the thermal noise in the microphone element, shall possess a damping factor such that 0.2 second shall be required to produce full scale deflection upon application of a constant a-c voltage.

a. The microphone element shall be tested in a location sufficiently free from noise and vibration so that the microphone element output results only from disturbances generated within the microphone element.

b. The test shall consist of connecting the microphone element in the test circuit with the current flowing through the element. The microphone shall be rotated as specified in 4.4.6.1g and agitated for 10 seconds with approximately a 70-dyne per square centimeter sound pressure varying uniformly from 1000 to 3000 cps and back to 1000 cps every 2 seconds. Then with the agitation off, measure the a-c voltage across the load resistor created by thermal noise.

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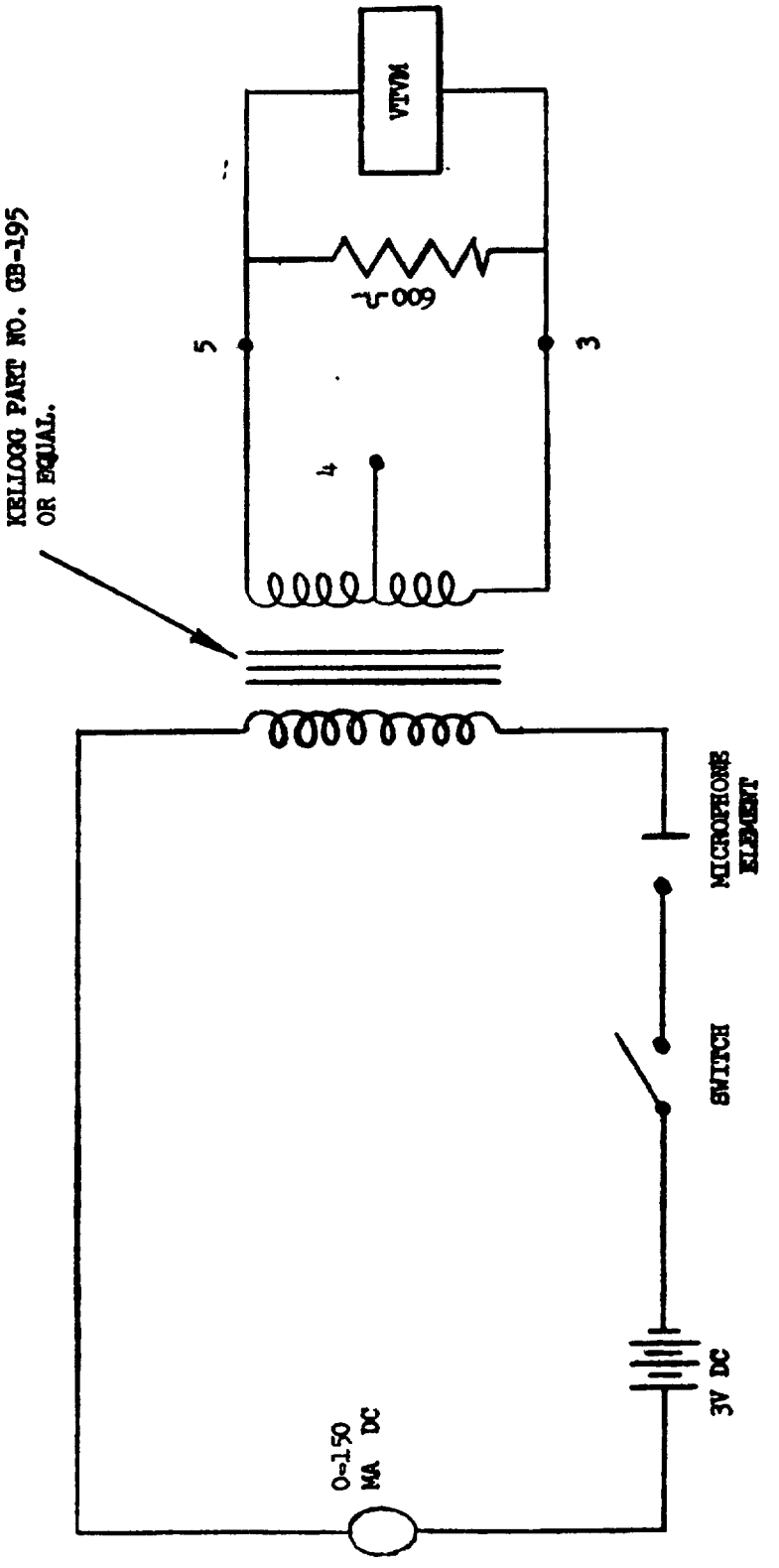


FIGURE 5. LOW VOLTAGE TEST CIRCUIT FOR MICROPHONE ELEMENT FREQUENCY RESPONSE

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c. The highest a-c voltage generated across the load resistance during the period from 5 seconds after agitation until the end of 2 minutes shall be taken as the thermal noise.

d. The resistance for the test shall be computed from the maximum reading of a d-c voltmeter connected across the microphone element during the period from 5 seconds after agitation, as specified in 4.4.6.5b, until the end of the test. The current reading shall be taken simultaneously.

e. The following limits of thermal noise shall apply to the tests as specified in 4.4.6.5 on individual microphone elements at any angle of microphone-element diaphram relative to the vertical. An aged microphone shall be defined as having been subjected to the vibration test.

(1) Thermal noise (new microphone element): 2.25×10^{-3} volts
(2) Thermal noise (aged microphone elements): 12.7×10^{-3} volts

f. The greatest resistance of a microphone element, obtained in this test, shall be not more than twice the greatest resistance of the same microphone element prior to the vibration test.

4.4.7 RELIABILITY TEST.- The equipment shall be operated continuously under ambient conditions for a period of 500 hours. During this test the voltage, current, loading, and method of excitation shall, insofar as practical, correspond to actual field conditions. Proper instrumentation shall be provided to insure immediate recognition of a failure. The occurrence of 2 or more failures shall be adquate cause for rejection. Such adjustments and realignments as are deemed necessary may be performed during the repair interval with the approval of the procuring activity.

4.5 PREPRODUCTION TESTING

4.5.1 PREPRODUCTION TEST SAMPLES.- The preproduction test samples shall consist of models representative of the production equipment. They shall be tested at a laboratory designated by the procuring activity or, when so stated in the contract, at the contractor's plant under the supervision of the procuring activity.

4.5.2 PREPRODUCTION TEST REPORT.- When the contractor performs the preproduction tests, he shall prepare a preproduction test report according to MIL-T-9107 and furnish three complete copies of the report to the procuring activity.

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4.5.3 TEST COUPLER.- The preproduction samples shall be accomplished by a 6 cc coupler, or equal, that shall be used for testing the receiver elements. Complete manufacturer's drawings of this 6 cc coupler shall also be submitted. The coupler and the drawings will be returned to the contractor upon approval of samples or termination of contract.

4.5.4 PREPRODUCTION TESTS.- Preproduction tests shall consist of all tests described under "4.4 TEST METHODS".

5. PREPARATION FOR DELIVERY

5.1 PREPARATION FOR SHIPMENT.- The headset-microphone shall be prepared and marked for shipment in accordance with MIL-P-17555. The shipment marking nomenclature shall be as follows: "Headset-Microphone 163( )/U".

6. NOTES

6.1 INTENDED USE.- The Headset-Microphone 163( )/U covered by this specification is a headset-microphone combination intended for use in military aircraft for all altitudes up to 15,000 feet.

6.2 ORDERING DATA.- Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Whether the preproduction test will be conducted by the procuring activity or by the contractor. (See 4.5.1)

c. The level of preservation, packaging and packing required. (See 5.1)

d. Conditions under which the reliability prediction report will be submitted for approval. (See 3.4.4.1)

6.3 DRAWINGS.- Upon the specific request of the contractor to the procuring activity, a copy of the applicable drawings of the equipment may be furnished, if available, for informational purposes only.

6.4 NOMENCLATURE.- The parentheses appearing in the nomenclature will be dropped or replaced by a letter, to indicate the particular design. Nomenclature for the individual components, where required, will be furnished upon application to the procuring activity.

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NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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MIL-H-9697(USAF)
AMENDMENT 1
30 September 1958

MILITARY SPECIFICATION

HEADSET-MICROPHONE H-163( )/U

This amendment forms a part of Military Specification
MIL-H-9697(USAF) dated 6 August 1958

Page 2, paragraph 3.2, item 1: Delete "Housing" and substitute "Assembly".
Page 2, paragraph 3.2, item 2: Delete "Housing" and substitute "Assembly".
Page 3, paragraph 3.4: Delete in its entirety and substitute the following:

"3.4 DESIGN AND CONSTRUCTION.- The headset-microphone shall consist of a headband assembly and a microphone assembly. The headband assembly shall contain two receiver assemblies. Each receiver assembly shall consist of a receiver housing and a receiver element. Receiver housings shall be attached to the headband by means of a movable metal band with an adjustment block and a metal fork. The microphone assembly shall consist of a hollow square boom anchored inside one receiver housing on one end and having a transmitter housing anchored on the other end. The transmitter housing shall contain a transmitting element. The two receiver housings and the transmitter housing shall each have a screw cap for holding the elements in place. The square boom shall contain the wiring between the transmitter and receiver elements with the wiring connected to terminals in the transmitter and receiver housing".

Page 3, paragraph 3.4.1, line 1: Delete and substitute:

"3.4.1 RECEIVER ELEMENTS.- The headset-microphone receiver elements shall be Western Electric Co. Type HC-3, or equal".

Page 3, paragraph 3.5.1: Delete "RECEIVERS"and substitute "RECEIVER".
Page 3, paragraph 3.5.1.1, line 1: Delete "receivers" and substitute "receiver".
Page 4, paragraph 3.5.1.3, line 1: Delete "receivers" and substitute "receiver".
Page 4, paragraph 3.5.1.4, line 1: Delete "receivers" and substitute "receiver".
Page 8, paragraph 3.6.1: Delete in its entirety and substitute the following:

"3.6.1 RECEIVER ASSEMBLY.- The receiver assembly shall consist of a receiver housing and a receiver element. There shall be a snug fit between the circular slot in the receiver housing and the rotating headband assembly ring to permit the microphone to remain in a desired set position."

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Page 9, paragraph 3.6.4, line 4: Delete "follows" and substitute "shown in figure 6".
Page 9, paragraph 3.6.4: Delete subparagraphs "a", "b" and "c" in their entirety.
Page 10, paragraph 4.4.2.2: Delete in its entirety.
Page 12, paragraph 4.4.5, subparagraph "a", line 5: After "element" add "or".
Page 12, paragraph 4.4.5 subparagraph "d": Delete in its entirety and substitute the following: "d". Over-voltage: After applying 10 volts at 400 cps directly across the receiver element terminals for 10 hours, the response shall be as specified in 3.5.1.4".
Page 18: Add the following new paragraph:

"4.4.8 RECEIVER HOUSING METAL BAND TEST.- The movable metal band on the receiver housing shall be given a reciprocating motion test through a 60° arc. Duration of the test shall be 10,000 cycles. At the completion of the test the microphone shall remain in any desired set position."

Page 19, paragraph 6.1, line 3: Delete "aircraft for all altitudes up to 15,000 feet". and substitute "ground communications".

RDP (RADC)
RCSSES

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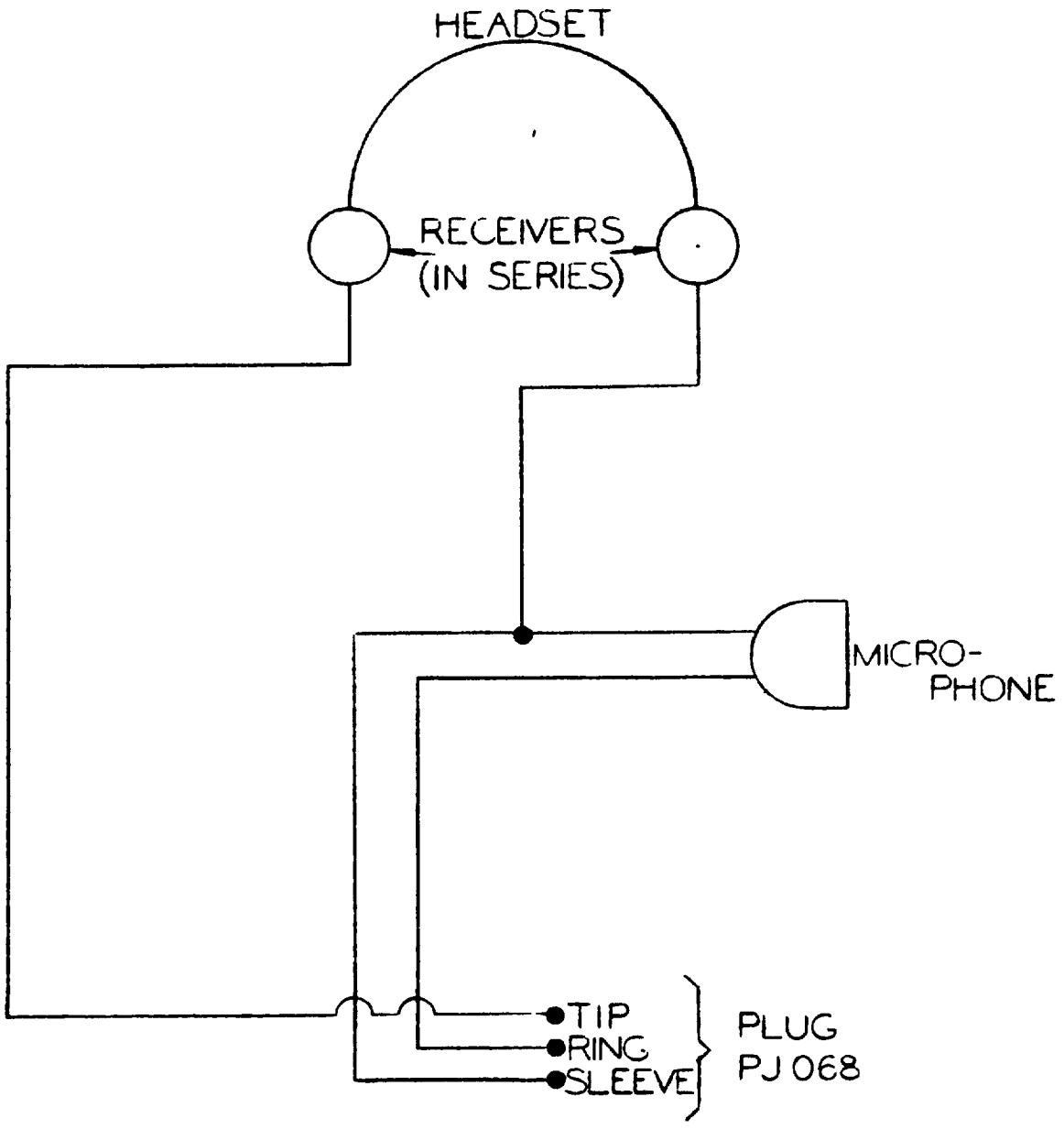


FIGURE 6. WIRING OF HEADSET-MICROPHONE

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